

Nicholas Wong

nicwjh@mit.edu — nicwong.com — [LinkedIn](#)

EDUCATION

Massachusetts Institute of Technology	2026 MFin , Master of Finance GPA: 4.6/5.0
UNC Chapel Hill	2024 B.S. , Computer Science; B.S. , Economics; Minor , Statistics and Analytics GPA: 3.95/4.0; Phi Beta Kappa Graduated with Highest Distinction
Stanford University	2026–28 Visiting Research Fellow (Non-Degree)
Harvard University	2024 Cross-Registration (Non-Degree)

Selected Coursework: Real Analysis; Probability Theory; Linear Algebra; Differential Equations; Optimization; Advanced Econometrics; Time Series Analysis; Machine Learning; Deep Learning; Advanced Financial Mathematics; Fixed Income & Derivatives; Options & Futures; Data Structures; Algorithms & Analysis; Files & Databases

RESEARCH EXPERIENCE

I am broadly interested in computational and statistical methods for problems in economics and finance. My work spans optimization, statistical learning, causal inference, and forecasting, with applications to financial markets, policy evaluation, and economic decision-making.

Stanford University Graduate School of Business (Palo Alto, CA, USA)	2026–present	Incoming Research Fellow Supervisor: Tim de Silva
Massachusetts Institute of Technology Sloan School of Management (Cambridge, MA, USA)	2026	Research Assistant Supervisor: Haoxiang Zhu

Treasury Market Microstructure

- Analyzing transaction-level US Treasury data to measure and characterize liquidity in the secondary Treasury market.

Verition Fund Management | 2025
Fixed Income & Macro Group
(New York, NY, USA)

Intern, Macroeconomic Research
Supervisor: Nisarg Kamdar (Portfolio Manager)

A Quantitative Framework for Macroeconomic Forecasting

- Built a real-time machine learning system to forecast U.S. Nonfarm Payrolls surprises using 20+ years of data; achieved statistically significant predictive power and generated a measurable trading edge. (Python)
- Designed calibrated uncertainty bands using four statistical models (density estimation), including a novel dynamic multiplier that widens forecasts under elevated disagreement; matched target confidence levels within 1–2% and enabled more accurate risk and volatility assessment.

MIT–Panagora Asset Management | 2025
MIT Sloan Finance Lab
(Boston, MA, USA)

Finance Research Practicum
Supervisors: Bhushan Vartak & Gita Rao

Large Language Models in Equity Research

- Built a rolling-window XGBoost classification framework to predict earnings changes and surprises for Russell 3000 stocks using 153 asset pricing factors, financial statement data, and LLM-generated embeddings (direction, magnitude, confidence, narrative text embeddings from three open-source models). Found that LLM features improved classification accuracy most when paired with financial statement inputs, suggesting LLMs provide complementary narrative context beyond raw numerical data.
- Constructed long-short tercile portfolios from XGBoost prediction signals, demonstrating that LLM-augmented strategies achieved up to 20% annualized alpha with information ratios above 1.0, significantly outperforming both base factor models and consensus analyst forecasts across multiple lag specifications.
- Implemented chain-of-thought prompting pipelines across three open-source LLMs (Llama-3-8B, DeepSeek-R1-Distill variants) to generate earnings predictions and financial narrative analyses from anonymized Compustat balance sheets and income statements, contributing to prompt design, data pipeline construction, and performance evaluation against IBES analyst benchmarks. (Python, XGBoost, HuggingFace)
- Offered quantitative research summer 2025 internship opportunity at PanAgora Asset Management (equity investment team, stock selector equity strategies)
- Poster presented at MIT Finance Lab Poster Session (March 2025)

MIT–T. Rowe Price | 2024
MIT Sloan Proseminar in Capital Markets
(Cambridge, MA, USA)

Research Practicum
Supervisors: Mark Kritzman & Sébastien Page (CIO)

Artificial Intelligence for Financial Analysis

- Designed a three-layered empirical framework with escalating prompt complexity to evaluate GPT-4o's ability to generate sell-side equity research reports for pure-play and multi-segment companies.
- Conducted a literature review synthesizing recent findings on LLM performance in quantitative financial analysis, including chain-of-thought prompting, earnings prediction, and human-AI complementarity.

- Benchmarked AI-generated reports against human analyst reports using a rubric-based utility evaluation and NLP methods (TF-IDF keyword extraction, cosine similarity), finding that human analysts consistently outperform across depth, accuracy, and actionability. (Python)

RESEARCH PROJECTS

- **Double Descent in Financial Time Series.** Nicholas Wong. Individual research project for *6.7960 Deep Learning*, Massachusetts Institute of Technology. Course supervised by Dr. Kaiming He, Dr. Sara Beery, Dr. Omar Khattab. Empirical study of generalization behavior in overparameterized neural networks applied to synthetic and financial time series data, examining the conditions under which double descent emerges and fails in low signal-to-noise environments.
- **Portfolio Optimization.** Nicholas Wong. Individual research project conducted at Harvard University for *Machine Learning and Big Data*, supervised by Dr. Soroush Saghafian. Developed an ensemble machine-learning-based forecasting and sparsified Markowitz optimization framework for NASDAQ-100 equities, achieving improved risk-adjusted performance relative to market benchmarks.
- **Regulatory Response to the SVB Collapse: A Natural Language Processing Analysis.** Nicholas Wong and Priscilla Clark. Research project conducted at the Massachusetts Institute of Technology for *Advanced Data Analytics & Machine Learning in Finance*; supervised by Dr. Mike Chen and Dr. Andrew Zachary. Applied natural language processing techniques to analyze shifts in U.S. banking regulatory discourse before and after the Silicon Valley Bank collapse, testing research hypotheses and documenting increased emphasis on liquidity, capital adequacy, and systemic risk oversight.
- **Evaluating CFTC COT Data as a Predictor of Treasury-Futures Returns.** Nicholas Wong. Individual research project as part of *15.437 Options & Futures Markets*, Massachusetts Institute of Technology. Course supervised by Dr. Haoxiang Zhu. Tested whether weekly CFTC Commitments of Traders positions predict short-horizon returns on 2-year and 10-year U.S. Treasury futures using publication-lagged signals, HAC-corrected predictive regressions, expanding-window out-of-sample forecasts, and sign-based trading strategies. Found no statistical or economic forecasting power across all specifications, consistent with the rapid incorporation of public position disclosures into liquid Treasury markets.

TEACHING EXPERIENCE

Massachusetts Institute of Technology MIT EECS (Cambridge, MA, USA)	2025	Laboratory Assistant Intro. to Programming and Computer Science
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- Held weekly office hours and coached students on Python programming fundamentals, debugging techniques, and best practices; assisted in course delivery by reviewing problem sets and model solutions prior to release.

AWARDS

- Phi Beta Kappa, 2023.

- Dean's List (multiple awards), 2021-2023.
- Merit Scholarship, New York University.

LEADERSHIP

- Former Reconnaissance Team Leader, Singapore Armed Forces; led reconnaissance platoon of 30 soldiers in combat training and exercises.

MISCELLANY

- Technical Skills: Python (Pandas, Polars, NumPy, Scikit-learn, Keras, TensorFlow, PyTorch), R, Stata, SQL, Java; Git; Linux/Unix; LaTeX; Bloomberg Terminal.
- Certifications: Machine Learning Specialization (Stanford University), Data Science Credential in Economics (UNC Chapel Hill), AI Fluency: Frameworks and Foundations (Anthropic), ChatGPT Prompt Engineering for Developers (DeepLearning.AI, OpenAI).
- Standardized Test Scores: GRE Quantitative Reasoning 169/170; SAT 1540/1600
- Activities: former competitive strength athlete (powerlifting) and competitive badminton player.
- Interests: fitness, barbell sports, health & wellness, cooking, history.